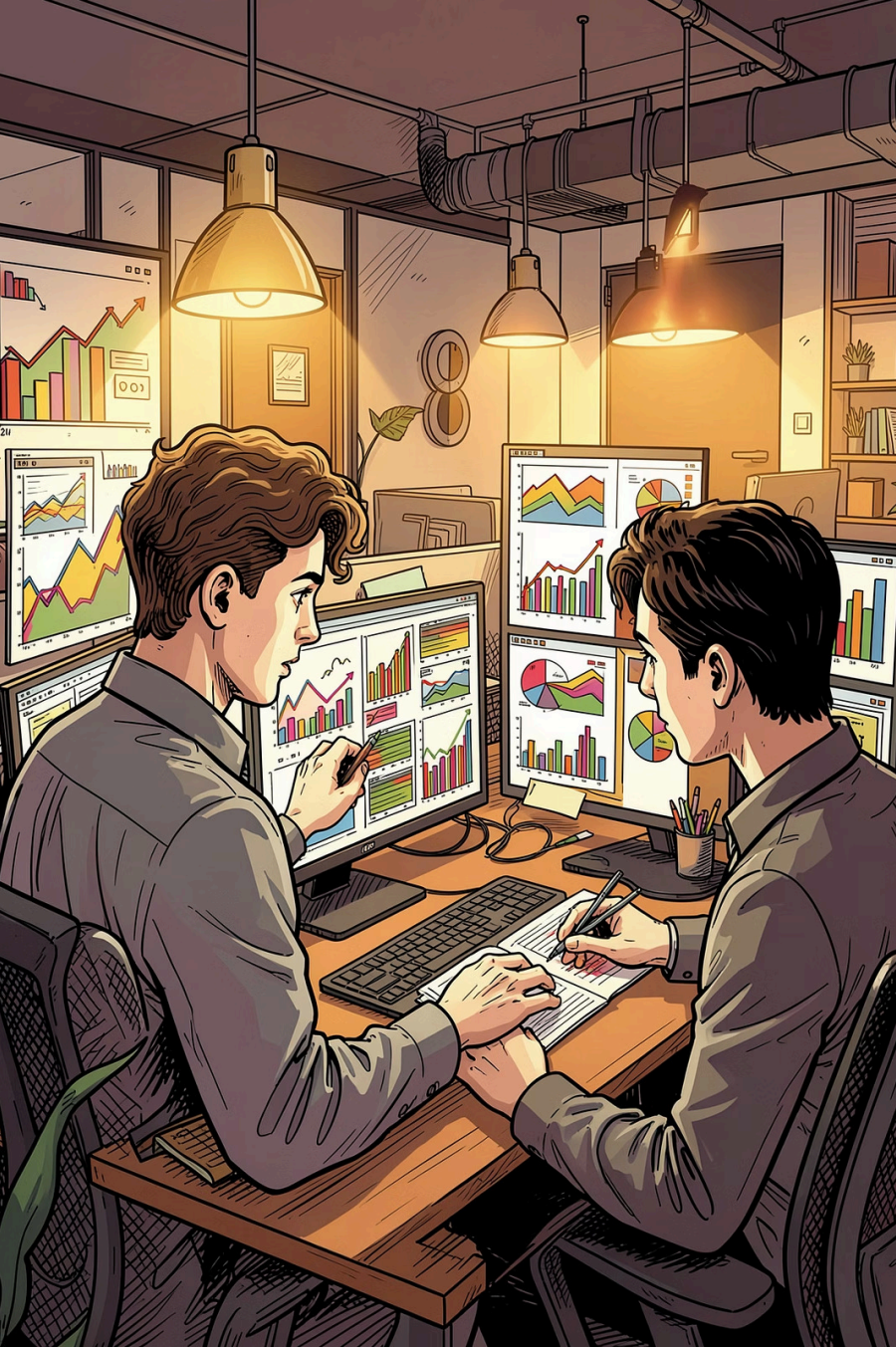




Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.

Dataset at a Glance

3,900

Purchases

18

Columns

37

Missing Ratings

Key Features

- **Demographics:** Age, Gender, Location, Subscription
- **Purchase:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Ratings, Shipping

Data Preparation in Python

01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Filled missing Review Ratings with per-category median

03

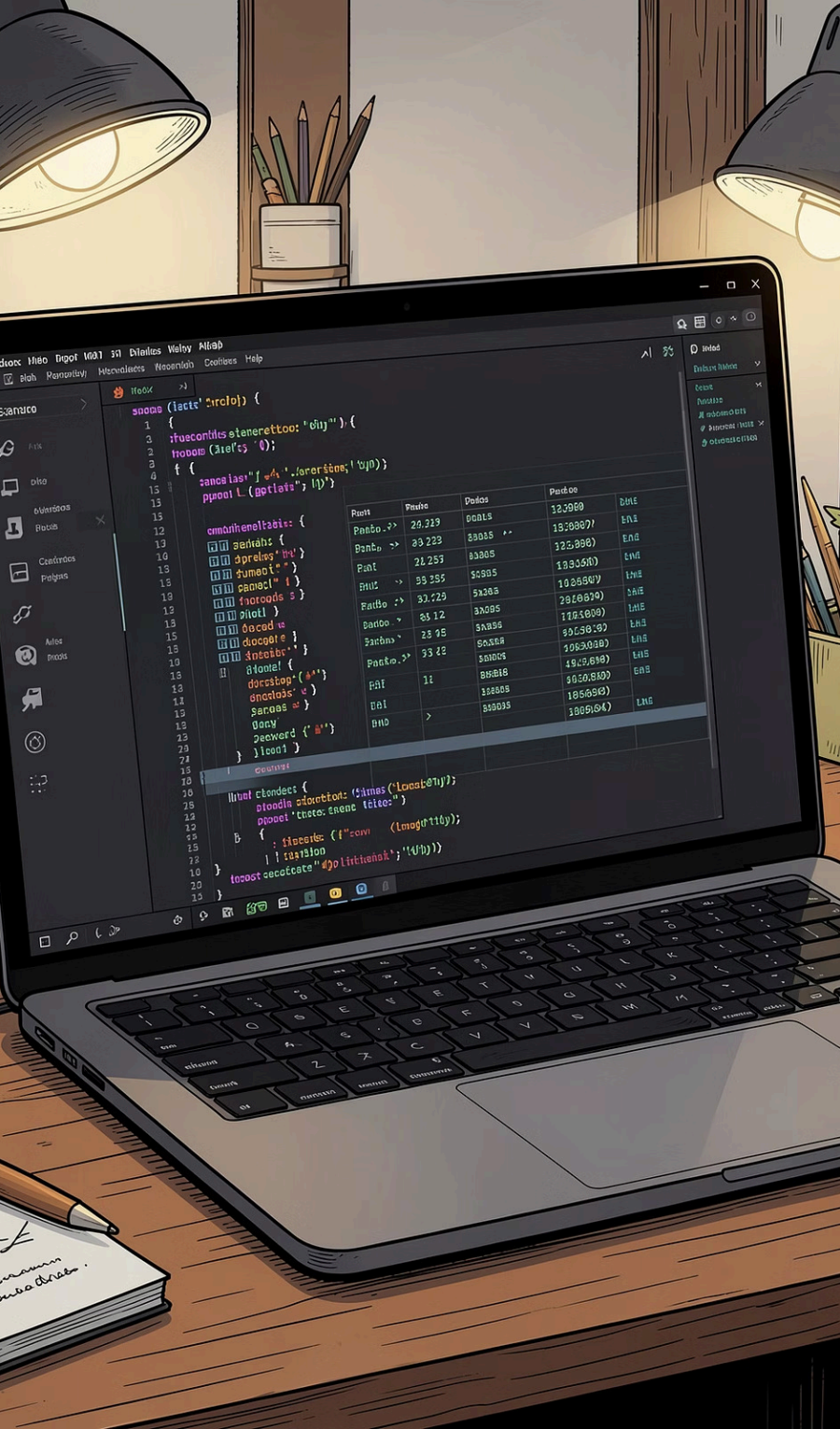
Engineer Features

Created `age_group` and `purchase_frequency_days` columns

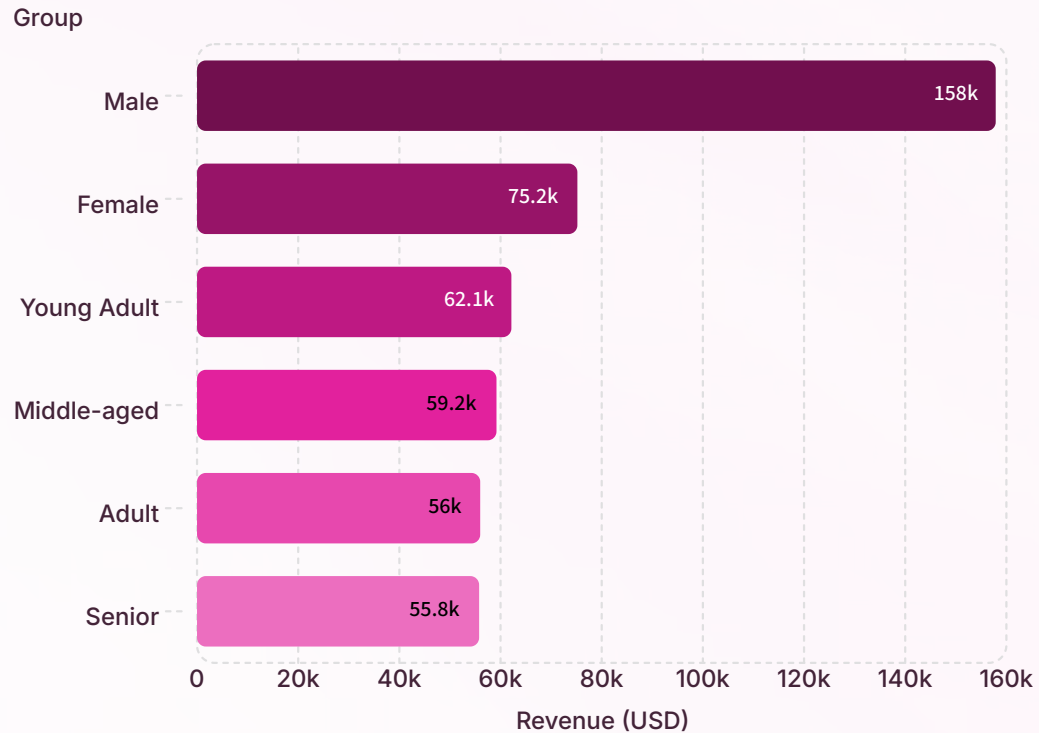
04

Load to PostgreSQL

Standardized to `snake_case`; dropped redundant `promo_code_used`



Revenue Insights



Key Takeaways

- Male customers generate **2× more revenue** than female
- **Young Adults** are the top revenue age group at \$62,143
- Revenue is relatively balanced across adult age groups

Subscription & Discount Behavior

Subscribers vs. Non-Subscribers

Status	Customers	Avg Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

 Only **27%** are subscribers — a major growth opportunity.

High-Spending Discount Users

839 customers used discounts yet spent *above* the average purchase amount.

Most Discount-Dependent Products

- Hat — **50%** discounted
- Sneakers — 49.66%
- Coat — 49.07%
- Sweater — 48.17%
- Pants — 47.37%

Top Products & Ratings

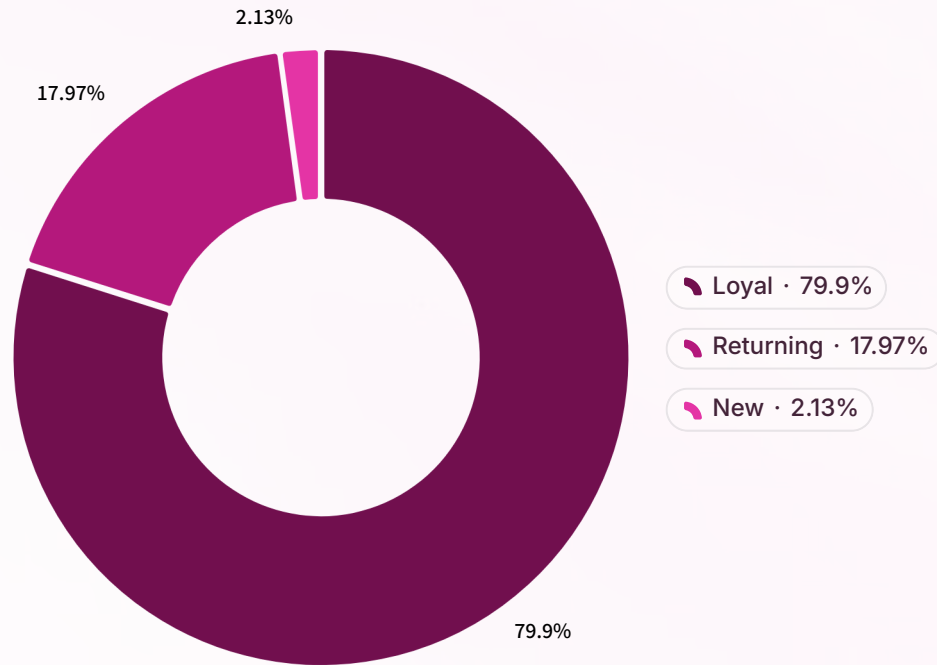
Top 5 by Avg Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Top 3 per Category

Category	#1	#2	#3
Accessories	Jewelry	Sunglasses	Belt
Clothing	Blouse	Pants	Shirt
Footwear	Sandals	Shoes	Sneakers
Outerwear	Jacket	Coat	—

Customer Segmentation



Segment Breakdown

Loyal — 3,116

More than **5 purchases**;
958 are also subscribers

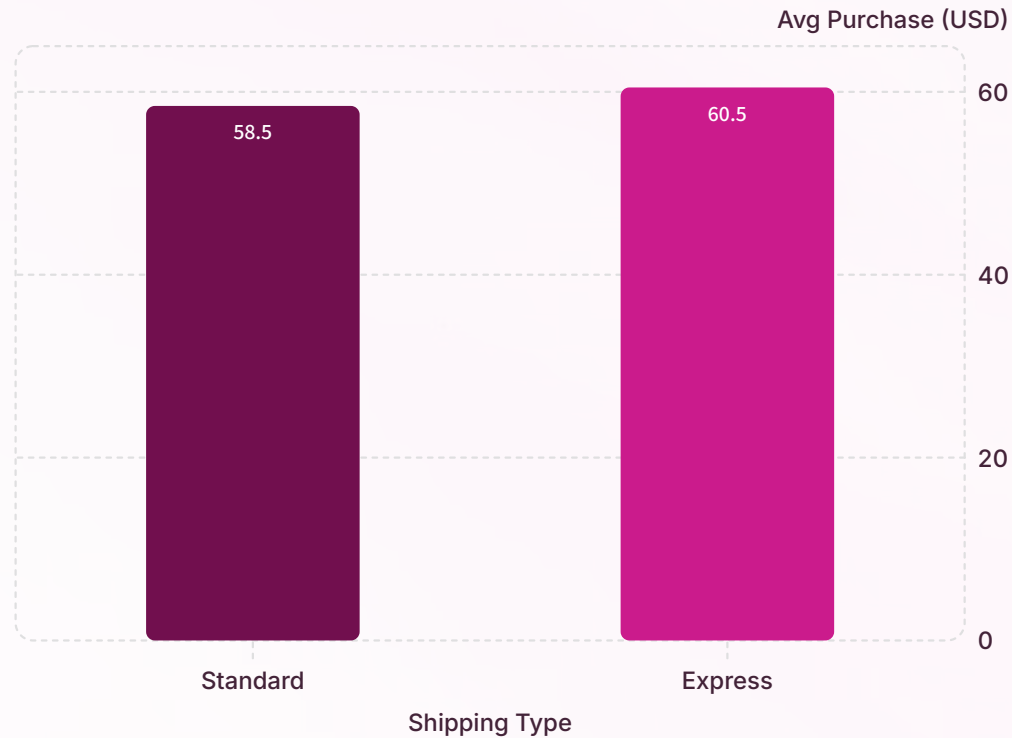
Returning — 701

Repeat buyers not yet in
loyal tier

New — 83

First-time purchasers with high conversion potential

Shipping & Spending Patterns



What This Tells Us

Express shipping users spend **\$2 more on average** — a signal of higher purchase intent.



Target express-shipping customers with premium product campaigns.

Power BI Dashboard



3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

27% / 73%

Subscribers vs. Non

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups and express-shipping users



Product Positioning

Highlight top-rated and best-selling items in campaigns